Introduction to Vectors

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Vectors are combined linearly; the dot product supplies length, angle, and orthogonality; matrices encode linear combinations and linear equations; independence and invertibility connect geometry with algebra.

Vectors and Linear Combinations

$$v + w = (v_1 + w_1, \ldots, v_n + w_n), \qquad cv = (cv_1, \ldots, cv_n).$$

Linear combination and span:
For $v_1, \ldots, v_n \in \mathbb{R}^m$ and $c_1, \ldots, c_n \in \mathbb{R}$, the vector $\sum_{j=1}^n c_j v_j$ is a linear combination. Its possible values form

$$\operatorname{span}\{v_1, \ldots, v_n\} = \Big\{\sum_{j=1}^n c_j v_j : c_j \in \mathbb{R}\Big\}.$$

Matrix form:
If $A = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix}$ and $x = (x_1, \ldots, x_n)^T$, then x_j is the coefficient of the jth column a_j:

$$Ax = \sum_{j=1}^n x_j a_j = b.$$

Linear dependence theorem:
$v_1, \ldots, v_n$ are dependent iff $\sum_{j=1}^n c_j v_j = 0$ for coefficients $c_j \in \mathbb{R}$ that are not all zero.

Dot Products and Geometry

$$v \cdot w = v^T w = \sum_{i=1}^n v_i w_i, \qquad \|v\| = \sqrt{v \cdot v}.$$

Cosine formula:
for nonzero v, w, $\cos\theta = \dfrac{v \cdot w}{\|v\|\|w\|}$.

Cauchy-Schwarz inequality:
$|v \cdot w| \le \|v\|\|w\|$.

Triangle inequality:
$\|v + w\| \le \|v\| + \|w\|$.

Pythagorean theorem:
$v \cdot w = 0 \Rightarrow \|v + w\|^2 = \|v\|^2 + \|w\|^2$.

$$\|v - w\|^2 = \|v\|^2 - 2v \cdot w + \|w\|^2.$$

Matrices and Equations

Invertibility criterion (square case):
A is invertible iff $Ax = 0$ has only $x = 0$ iff its columns are independent; then $x = A^{-1}b$.

$$AB = (Ab_1 \ \cdots \ Ab_p), \quad (AB)^T = B^T A^T, \quad A(I) = A.$$

PROBLEM SET

Q1.

Let $v = (v_1, v_2)$ and $w = (w_1, w_2)$ be nonzero vectors with $v_1 w_1 \neq 0$. Suppose the product of the slopes of the lines joining the origin to v and w is -1. Show that $v \cdot w = 0$, and hence that the vectors are perpendicular.

Q2.

(i) Let $A \in \mathbb{R}^{m\times n}$ and $x \in \mathbb{R}^n$. For each i, let $r_i \in \mathbb{R}^n$ be the column vector whose transpose r_i^T is the ith row of A. If $Ax = 0$, show that $r_i \cdot x = 0$ for every i.

(ii) Hence explain why, when the rows span a plane in $\mathbb{R}^3$, that plane is perpendicular to x.

Q3.

(i) For

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix},$$

solve $Ax = b$ for an arbitrary $b = (b_1, b_2, b_3)^T$.

(ii) State the inverse matrix A^{-1}.

Q4.

(i) Find two different combinations of the three vectors $u = (1, 3)$, $v = (2, 7)$, and $w = (1, 5)$ that produce $b = (0, 1)$.

(ii) Show that, for any three vectors in the plane, if one combination produces b, then there are infinitely many distinct linear combinations that produce b.

Q5.

Let x, y, z be real numbers, not all zero, satisfying $x + y + z = 0$. Set $v = (x, y, z)$ and $w = (z, x, y)$. Find the angle between v and w by showing that

$$\frac{v \cdot w}{\|v\|\|w\|} = -\frac{1}{2}.$$

Q6.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a 2×2 real matrix. Prove that the rows of A are linearly dependent if and only if the columns of A are linearly dependent.

Q7.

(i) Show that there cannot be three nonzero, mutually orthogonal vectors in $\mathbb{R}^2$.

(ii) Show that there are at most n nonzero, mutually orthogonal vectors in $\mathbb{R}^n$.

(iii) *Optional.* Fix a small angle $\varepsilon > 0$. Make a conjecture about the growth, as n increases, of the maximum number of unit vectors that can be placed in $\mathbb{R}^n$ such that every pair forms an acute angle between $90° - \varepsilon$ and $90°$.

Q8.

(i) Let u and v be independent unit vectors chosen uniformly from the unit circle in $\mathbb{R}^2$. Find the mean value of $|u \cdot v|$.

(ii) *Optional.* Repeat the calculation for independent unit vectors chosen uniformly from the unit sphere in $\mathbb{R}^3$. Does the comparison support your conjecture in Q7(iii)?

Hint: If a continuous random variable X has probability density function f_X, then

$$\mathbb{E}[g(X)] = \int g(x) f_X(x)\, dx.$$

By rotational symmetry, one vector may be fixed; determine the distribution of the dot product using spherical surface area.